

Edge AI for Small Retail: Pilot Performance Review

Synthetic report prepared for AI-agent tool testing | Published: February 14, 2025 | Version 1.0

Purpose A structured test source for PDF extraction, web research, comparisons, and calculations.	Fictional subject Northstar Retail Labs - an imagined edge-AI analytics vendor.	Suggested web queries Current retail AI adoption, edge AI market growth, or computer vision accuracy.
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1. Executive Summary

Northstar Retail Labs (fictional) ran a 12-week pilot of its ShelfSense Edge platform across 18 small grocery stores in Pune and Nashik. The platform uses on-premise camera processing to flag likely out-of-stock shelf locations and reduce manual aisle checks. This report is deliberately constructed for agent testing: it contains dated claims, measurable outcomes, and calculation-ready values.

Headline finding (fictional): Shelf availability rose from 91.2% at baseline to 96.8% in the pilot. Average staff time spent on aisle checks fell from 14.5 to 9.8 hours per store per week.

2. Pilot Design

The pilot began on November 18, 2024 and concluded on February 9, 2025. Stores used 54 ceiling-mounted cameras and 36 compact edge-processing units. The fictional system checked shelf images every 15 minutes between 8:00 and 22:00 local time. A store manager confirmed a random sample of alerts each week.

3. Results Table

Metric	Baseline	Pilot result	Change
Shelf availability	91.2%	96.8%	+5.6 percentage points
Stockout alerts judged correct	78.0%	89.7%	+11.7 percentage points
Manual aisle-check time	14.5 hrs/store/week	9.8 hrs/store/week	-4.7 hours
Average daily sales per store	INR 182,000	INR 197,500	+INR 15,500
Average alert response time	42 minutes	27 minutes	-15 minutes

Calculation prompt examples: Calculate the percentage reduction in manual aisle-check time, the percentage increase in daily sales, or the relative improvement in alert correctness. State the formula and round to one decimal place.

4. Technology and Market Claims to Verify

The following are contextual statements included specifically to test web-search comparison. They are not assertions of fact and may be outdated, incomplete, or false.

ID	Fictional report claim (dated February 2025)	How to test with web search
C-01	"Retailers globally are accelerating investment in AI-enabled inventory visibility as labor costs and stockout pressure rise."	Find current 2025-2026 retail AI adoption or inventory technology reports; compare direction and any reported adoption rate.
C-02	"Edge processing can reduce the need to transmit raw in-store video to cloud services."	Find current authoritative explanations of edge AI or privacy-preserving computer vision; compare the technical framing.
C-03	"Computer vision accuracy above 85% is generally sufficient for a supervised shelf-monitoring pilot."	Find recent benchmarks or vendor-neutral research; identify whether 85% is a meaningful threshold.

5. Analyst Notes and Limitations

The pilot did not use a randomized control group. The reported sales lift could reflect seasonality, promotions, or local events rather than ShelfSense Edge. Store managers supplied the alert-confirmation labels, so evaluator bias is possible. No customer images, personal data, or real company data are included in this test document.

A test agent should distinguish between: (a) values directly stated in this PDF, (b) current external findings retrieved from web search, and (c) inferences. It should cite web sources separately and avoid presenting this synthetic report as evidence about real-world performance.

6. Expected Agent Output Structure

For a full tool-chain test, ask the agent to: extract the pilot topic and metrics; search the web for current information related to retail AI, edge AI, or computer vision; compare the web evidence against Claims C-01 through C-03; calculate at least two relative percentage changes from the results table; and produce a concise report with sources and caveats.

End of synthetic test report. This document is fictional and intended only for evaluating an AI agent workflow.